

Update on 2025 post-TS tracking efficiencies

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Tag-and-probe method

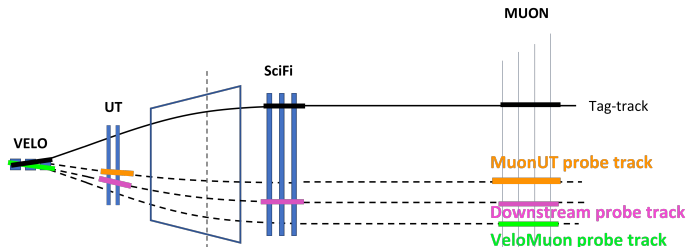


Figure by Rowina

- Tag-and-probe method with $J/\psi \rightarrow \mu^+ \mu^-$
- Fully reconstruct one muon...
- ... and partially reconstruct the other muon
- Match hits in specific sub-detector with partially reconstructed track

$$\epsilon_{\text{track}} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{failed}}}$$

Strategy for tracking efficiency determination:

- Combine efficiencies of two methods:
 - VeloMuon: SciFi efficiency
 - Downstream: Velo efficiency
 - Combined: $\text{Velo} \times \text{SciFi}$ efficiencies
- Cross check with long track efficiency from the MuonUT method
 - Main systematic

Recent presentations:

- [A/S week talk](#)
- [RTA WP4 talk on 2025 pre-TS corrections](#)
- For analysts: v0 tables available on GitLab [here](#)

Tracking efficiencies are determined using TrackCalib2

- GitLab repository [here](#)
- Mass fit done on tuples from AP to extract efficiencies
- Note: Analysts should not have to run TrackCalib2 themselves, unless the released numbers are not suitable (need different binning, etc)

Datasets considered:

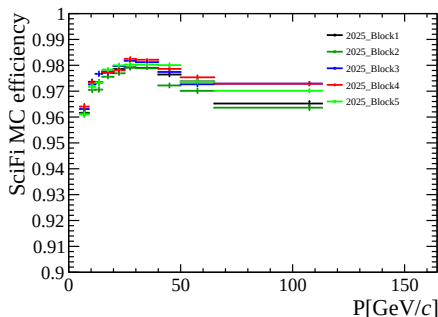
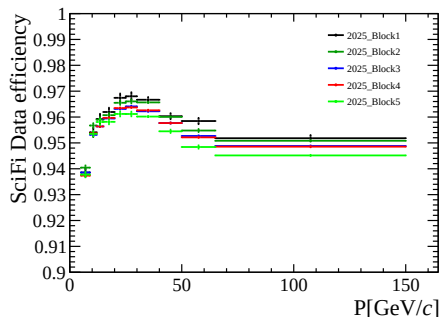
- Samples with Sim10g corrections available:
 - 2024 blocks 2, 1, 5, 6, 7, 8, *pp* ref
 - 2025 pre-TS (blocks 1, 2)
 - **2025 post-TS (blocks 3, 4, 5)**
- Samples not analysed yet due to missing MC:
 - 2024 blocks 4 and 3
 - 2026

Define 2025 blocks according to simulation conditions:

Block	Sprucing campaign	Magnet polarity	MC sample	$\mathcal{L}_{\text{int}}(\text{fb}^{-1})$
1	Sprucing25c1	MagDown	W21	0.4
2	Sprucing25c1	MagUp	W22.25	1.6
3	Sprucing25c2	MagDown	W28.36	4.3
4	Sprucing25c3/4	MagUp	W37.43	4.1
5	Sprucing5c4	MagDown	W43.45	1.0

2025 tracking efficiencies

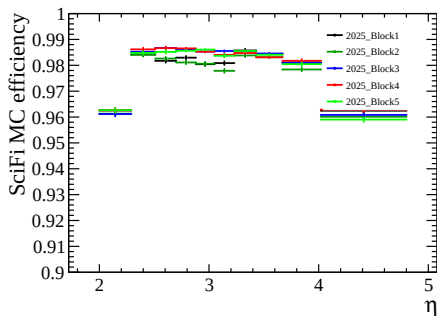
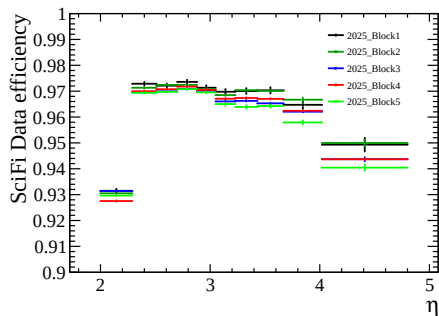
Comparison of tracking efficiencies in 2025 data blocks



SciFi efficiencies dropping slowly over time in data

2025 tracking efficiencies

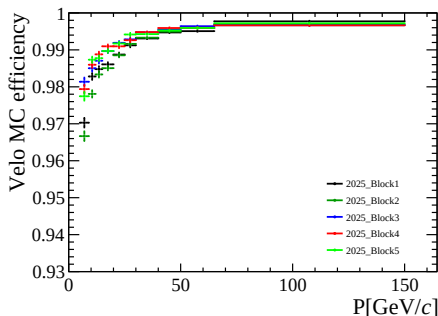
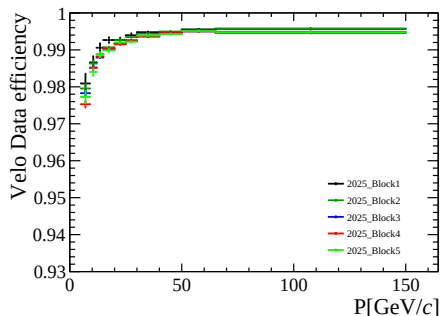
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2025 tracking efficiencies

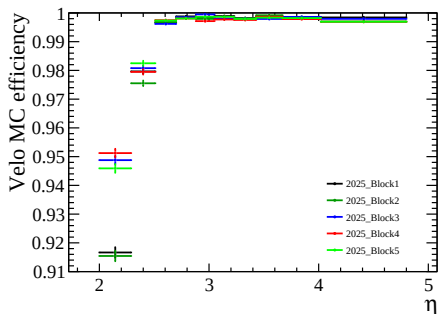
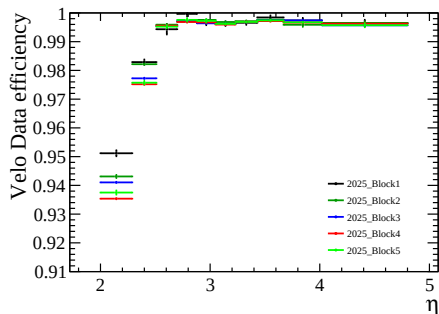
Comparison of tracking efficiencies in 2025 data blocks



Velo efficiencies underestimated in blocks 1/2, but not blocks 3/4/5

2025 tracking efficiencies

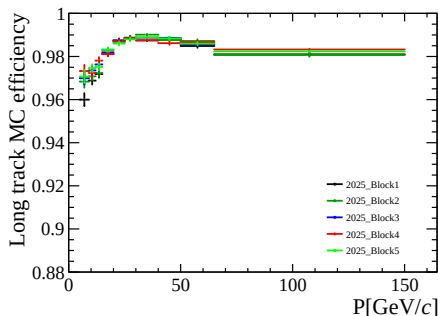
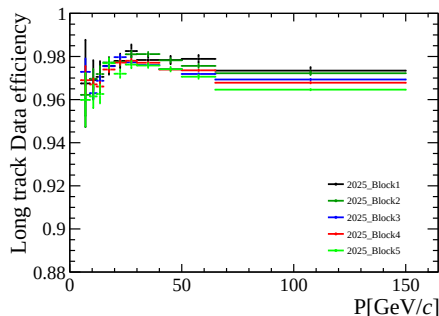
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2025 tracking efficiencies

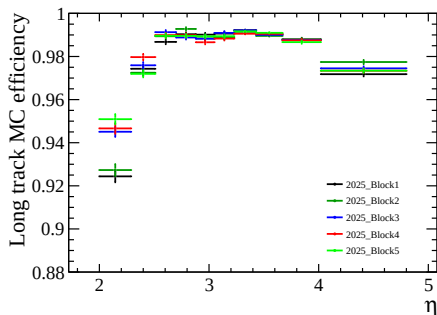
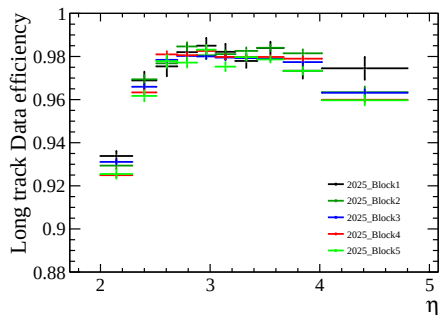
Comparison of tracking efficiencies in 2025 data blocks



MuonUT efficiencies look consistent, but lower statistics

2025 tracking efficiencies

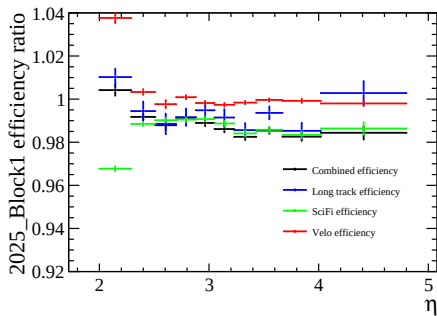
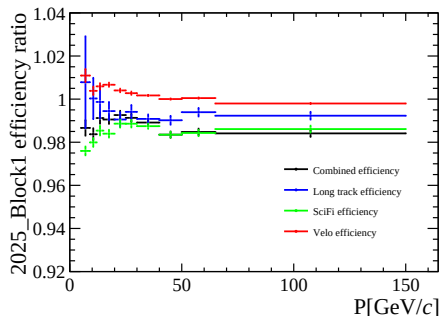
Comparison of tracking efficiencies in 2025 data blocks



MuonUT efficiencies look consistent, but lower statistics

2025 data/MC ratios in 1D

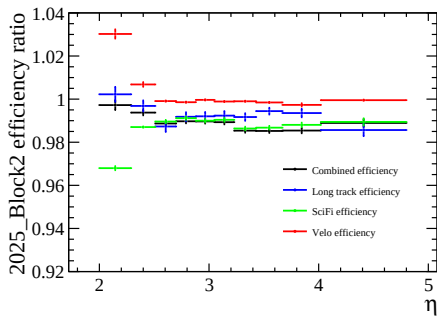
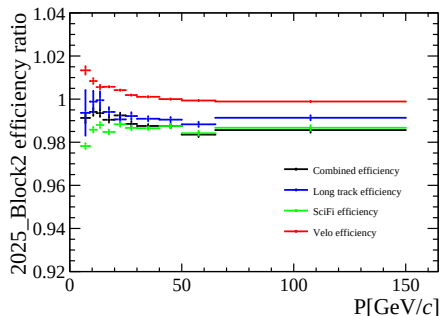
2025 pre-TS data/MC ratios, Block 1



Note that Velo efficiencies are underestimated for this block!

2025 data/MC ratios in 1D

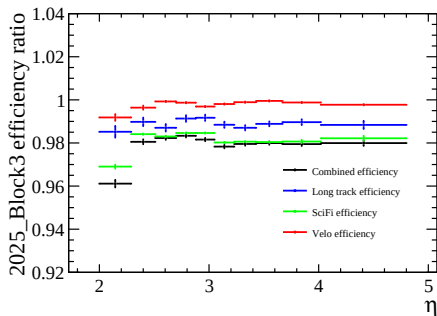
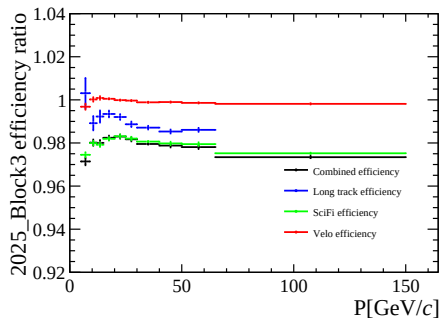
2025 pre-TS data/MC ratios, Block 2



Note that Velo efficiencies are underestimated for this block!

2025 data/MC ratios in 1D

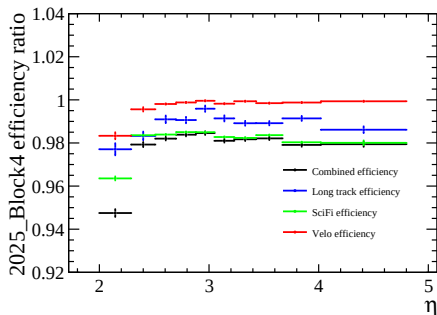
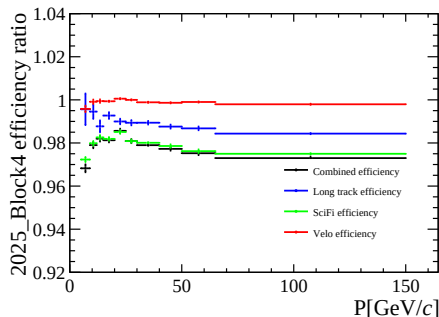
2025 post-TS data/MC ratios, Block 3



Much better Velo efficiency agreement

2025 data/MC ratios in 1D

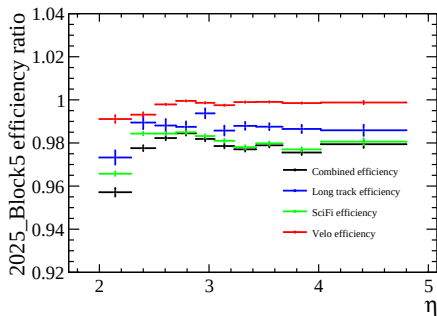
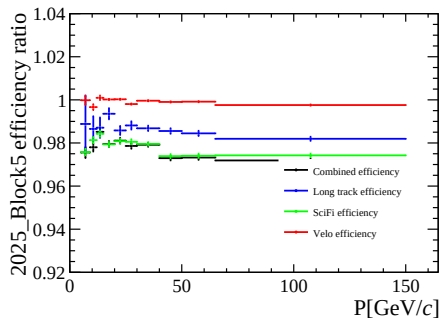
2025 post-TS data/MC ratios, Block 4



Much better Velo efficiency agreement

2025 data/MC ratios in 1D

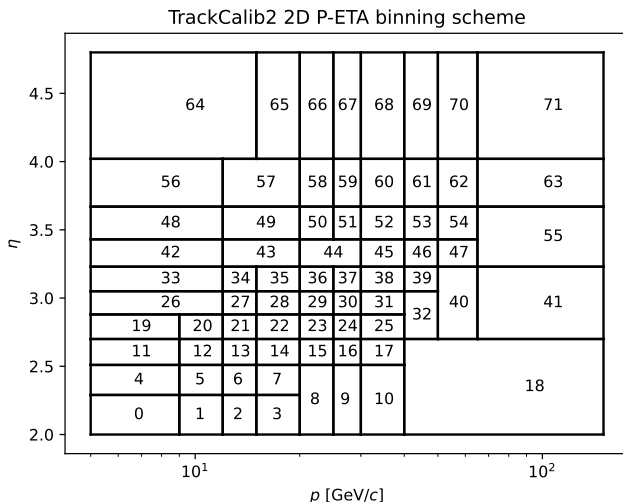
2025 post-TS data/MC ratios, Block 5

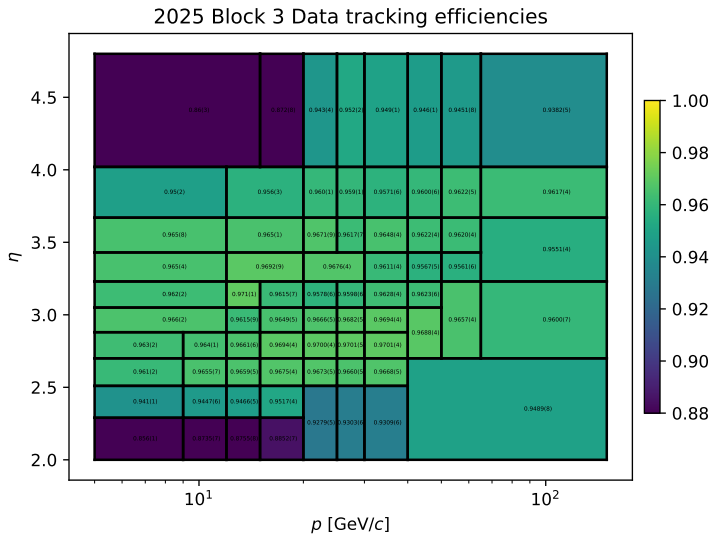


Much better Velo efficiency agreement

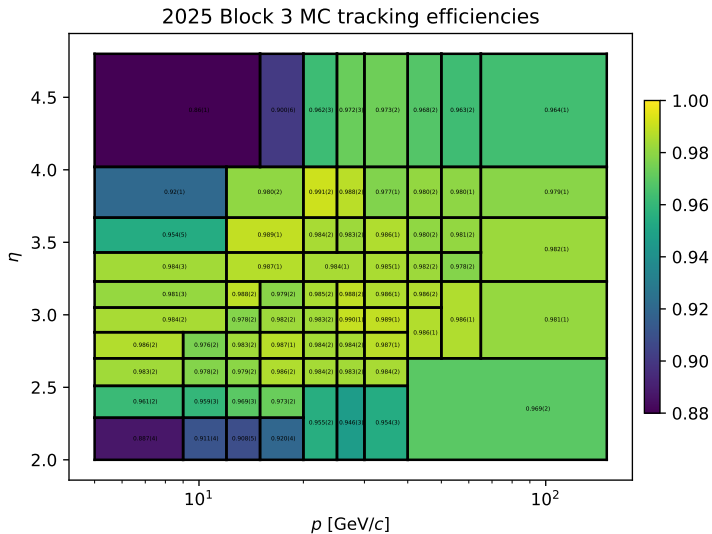
2D binning

2D binning in p vs η with 72 bins:

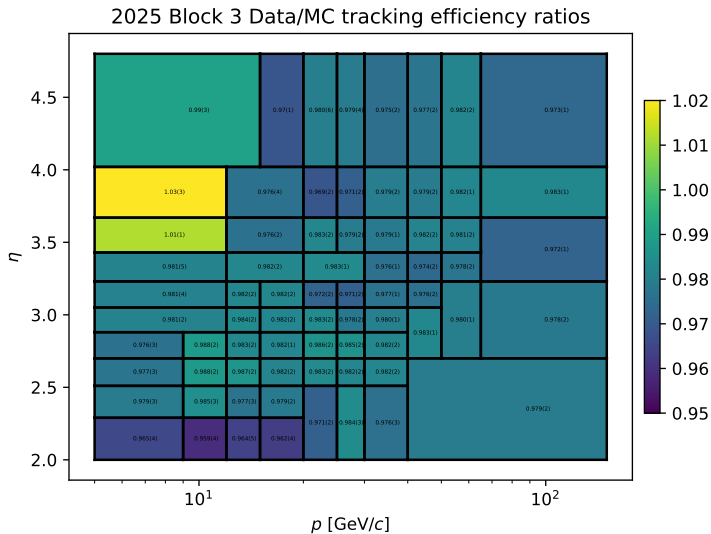




2D binning

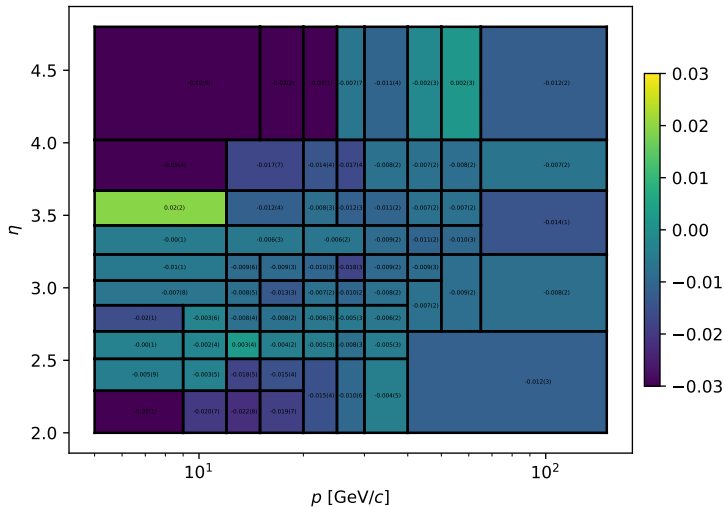


2D binning



2D binning

2025 Block 3 Data/MC Combined vs MuonUT ratio difference



We would like to release the tracking efficiency correction tables for 2025 post-TS

- 2024 and 2025 pres-TS v0 are already available on [GitLab](#)
- 2025 file will be updated with additional post-TS tables
- Feedback is welcome!

Some remarks on systematic uncertainties

Quick clarifications on systematic uncertainties:

- 1 Systematic provided is correlated between all bins, while statistical uncertainties are independent between bins
- 2 Currently systematics only consists of the Combined-MuonUT discrepancy, which is by far the largest systematic
- 3 At low p and η , corrections might vary significantly within each bin, and analyses that are sensitive to this effect should consider an additional systematic due to this

Summary:

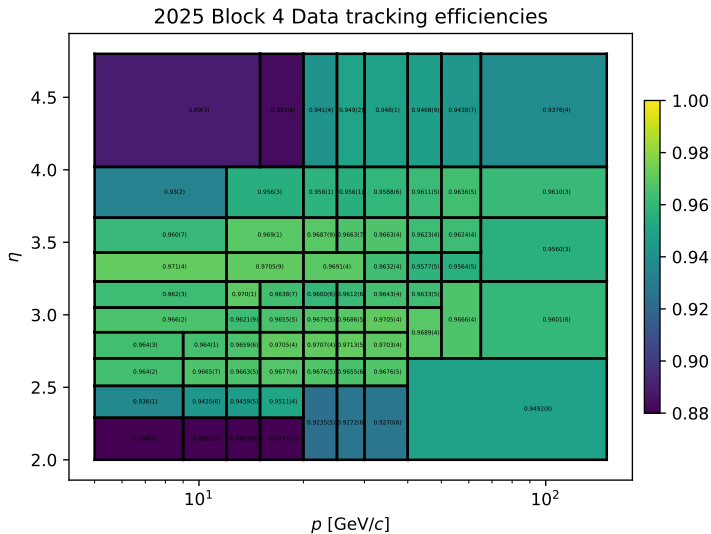
- 1 2025 post-TS tracking efficiencies show no issues
- 2 Correction tables for post-TS have been prepared

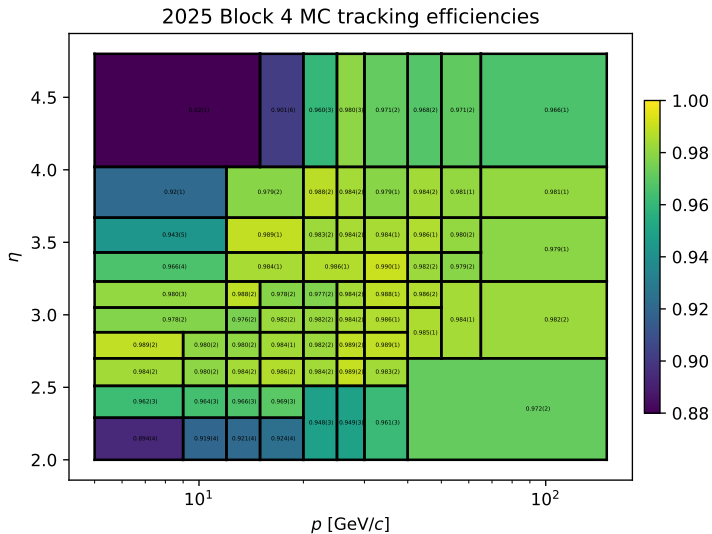
Next steps

- Finalise hadronic corrections
- Update tables using larger MC samples
 - Revised P-ETA binning
 - Separate tables for μ^+ and μ^-
 - Charge integrated table with left and right bins in ϕ

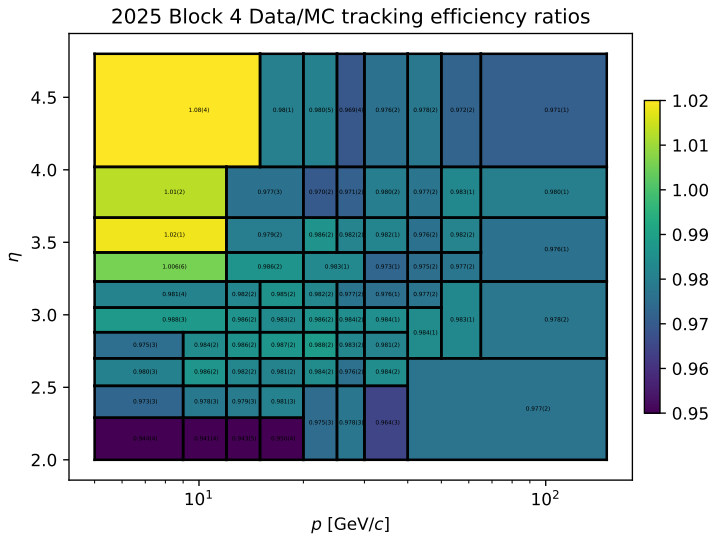
Thanks for your attention!

Backup

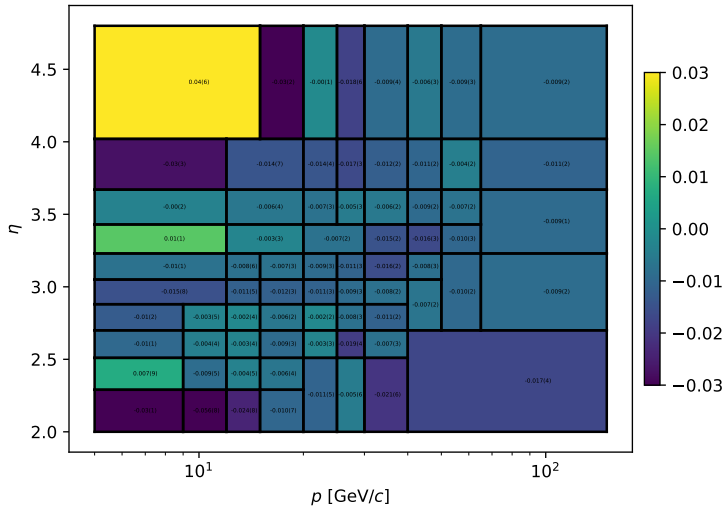


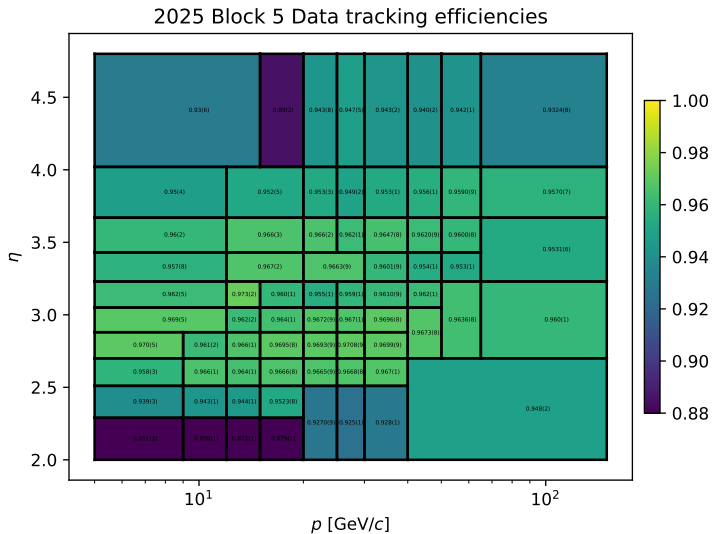


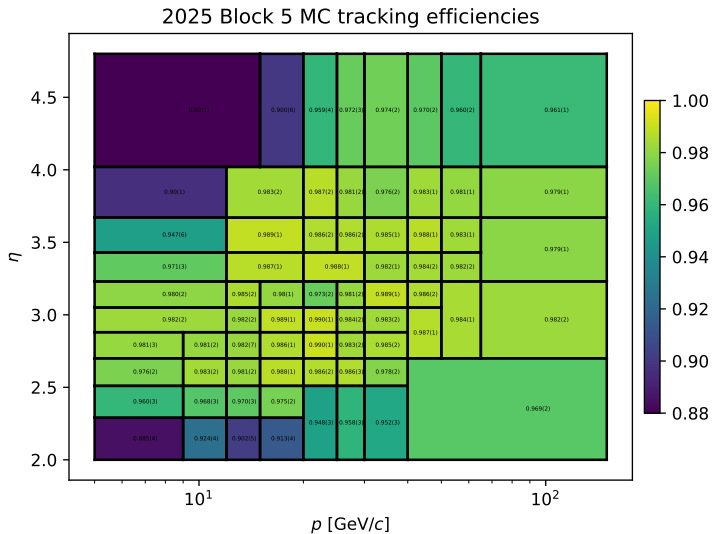
2D binning



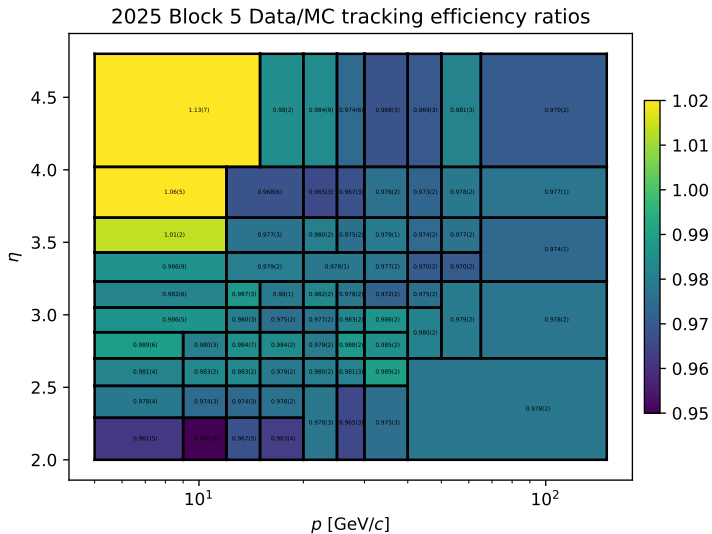
2025 Block 4 Data/MC Combined vs MuonUT ratio difference







2D binning



2D binning

2025 Block 5 Data/MC Combined vs MuonUT ratio difference

